ClinIQ

AI-Powered Clinical Assistant for Smart Patient History Intake and Real-Time Diagnosis Support

1. Problem Statement

Primary care is under strain in very different ways in different parts of the world, but the root problem is the same: doctors do not have enough time with each patient, and the history taking step that should come before diagnosis gets rushed or skipped entirely.

In Pakistan and much of South Asia, this shows up as a severe shortage of doctors relative to population, clinics that still run on paper records, and no assistant doctor available to prepare patients before a consultation. In the United States, the pressure looks different but leads to the same outcome: primary care shortages concentrated in rural areas and community health centers, shrinking visit times, and language barriers with patients who are not native English speakers. In both settings, the result is the same pattern of diagnostic error, missed symptoms, skipped follow up questions, and important history that never makes it into the record.

A few underlying issues repeat across both markets:

- **No fixed intake standard.** How thorough a history taking session is depends on how busy the clinic happens to be that day, not on a consistent clinical process, whether that clinic is in rural Punjab or rural Texas.
- **Language barriers.** Patients often describe symptoms in their own language or in everyday terms, whether that is Urdu, a regional language, or Spanish, and doctors have to translate this into clinical terms on the spot.
- **Records that are hard to use.** Encounters end up as paper notes, rushed typing, or systems that do not talk to each other, making past history slow to retrieve.
- **Staffing that smaller clinics cannot afford.** Rural health units in South Asia and rural or community clinics in the US both face the same constraint: they cannot hire extra clinical staff just to handle intake.

There is currently no accessible, doctor in the loop tool that automates structured history taking and prepares clinical reasoning before the physician walks into the room, in either a low resource setting or a high cost, understaffed one.

2. Objectives

ClinIQ is built to close this gap in both settings by acting as an AI assistant that prepares every patient encounter before the physician sees the patient. The main objectives are:

- **Automate structured history taking.** Run a consistent, conversational intake with every patient so that every important question about symptoms, duration, severity, medical history, medications and risk factors gets asked every time, regardless of how busy the clinic is.
- **Support clinical decision making, not replace it.** Turn the patient's everyday description into precise clinical terminology and produce a ranked differential diagnosis list, along with suggested investigations, positioned clearly as decision support that the physician reviews and decides on.
- **Digitize patient records automatically.** Turn every conversation into a structured, searchable record, removing the need for manual paperwork and making it easy to export into whatever system a clinic already uses.
- **Support multilingual patient interaction.** Let patients speak naturally in English, Spanish, Urdu or other regional languages, with automatic translation into standardized clinical terminology the physician can act on.
- **Work in both low resource and high cost settings.** Keep the core product lightweight enough to deploy in a rural health unit with limited infrastructure, while also being able to meet the compliance expectations of a US clinic.
- **Handle patient data responsibly everywhere.** Treat every conversation as sensitive health in-

formation, with security and privacy practices designed to meet HIPAA requirements in the US and equivalent data protection expectations elsewhere.

3. Business Use Case

ClinIQ is designed as a clinic facing SaaS product, not a consumer app, in both markets. Adoption is driven by physicians and clinic administrators, not by individual patients.

Target Customers

In Pakistan and South Asia: private clinics, hospital OPDs, and rural health units facing high patient loads with no or few assistant doctors. In the United States: rural primary care clinics, community health centers, urgent care clinics, and small primary care practices facing physician shortages and limited support staff.

Value Delivered

- Cuts down per patient time spent on history taking, in both high volume and understaffed settings.
- Reduces the risk of missed symptoms and diagnostic error through consistent, structured intake.
- Produces structured digital records automatically, without either kind of clinic hiring additional staff.
- Improves care access for patients who do not speak the clinic's primary language.

Revenue Model

- A tiered SaaS subscription per clinic, scaled by doctor, provider, or patient volume, priced according to each market.
- A per encounter pricing option for low volume rural clinics in either region.
- A free or low cost tier for solo practitioners, meant to drive early adoption in both markets.

Future Revenue Streams

Anonymized epidemiological and population health analytics for public health bodies, payers, and pharmaceutical companies, lab and pharmacy integrations, insurance pre authorization support in Pakistan, and value based care reporting support in the US.

Go to Market Strategy

Run parallel pilots: one with a private clinic chain or hospital OPD network in a single Pakistani city, and one with a community health center or small regional network in the US. Use measured consultation time savings and diagnostic accuracy from both pilots as the core evidence before scaling either market further.

Regulatory Considerations

Requirements differ sharply between the two markets. In the US, generating diagnostic suggestions likely means positioning ClinIQ as clinical decision support software under FDA guidance, with strict HIPAA compliant data handling. In Pakistan, the regulatory environment for this kind of software is comparatively lighter today, though data privacy practices should still be built in from the start rather than added later. The plan is to launch in the lower barrier market first and treat the other as a longer term expansion once the product and evidence base are strong enough to support it.

Market Opportunity

Pakistan has roughly one doctor per 1,000 people and hundreds of thousands of clinics still running on paper records. The United States faces a persistent primary care physician shortage concentrated in rural and underserved communities. Together these represent two very different but equally real markets for the same underlying product: one defined by scarcity of clinical staff, the other by high cost and limited time per patient.